Nopphrakorn Kerdsamut (Nop)

linkedin.com/in/nopphrakorn-kerdsamut
github.com/nopphrakornk

Email : nopphrakorn.k@gmail.com
Mobile : (+66) 8 5550 1122

Experience

- **PTTEP** — Bangkok, Thailand
  Data Scientist — *December '21 - Present*
  - **Production API**: Project and vendor management for model services.
  - **Search**: Currently working on search. TBD.

- **Kaidee** — Bangkok, Thailand
  Data Scientist — *July '20 - March '21*
  - **Post Page Suggestion API**: Built and deployed an api for suggestions on the listing page. The api contains three main models: *Atrributes-to-Price Regressor*, *Title-to-Category Classifier* and *Title-to-Price Regressor*. All models in the api are updated and retrained weekly.
    - **Attribute-to-Price Regressor**: Built a pipeline of autoencoder and multilayer perceptron model. The autoencoder scores the uniqueness of a vehicle given attributes where the multilayer perceptron suggests the price of the vehicle given encoded attributes.
    - **Title-to-Category Classifier**: CNN classifier to predict the category given ad listing text. Created a more streamlined listing experience.
    - **Title-to-Price Regressor**: CNN regressor to suggest 3 prices given ad listing text. The model is train with quantile loss to suggest price bounds.
  - **Search Term Extraction**: CNN model that predicts the top ten search terms given ad listing's title and description. Aim to improve SEO.

Technical Skills

- **Technical**: Python (Pandas, PyTorch, Scikit-Learn, PySpark), SQL, Hadoop, LaTeX
- **Languages**: Thai, English, Mandarin (aka 黃俊城)

Education

- **New York University** — New York City, NY
  M.S. in Data Science — *Sept 2018 – May 2020*
  - **Coursework:**: Machine Learning, Big Data, Computer Vision, Time Series Analysis, NLU, Text as Data

- **Mahidol University International College** — Nakhon Pathom, Thailand
  B.S. in Physics with Honors — *Apr. 2014 – Apr. 2018*
  - **Exchange Program**: University of North Carolina at Chapel Hill (Fall 2016)

Research and Selected Projects

- **Understood.org-CDS Datathon**: Implemented recommendation systems to predict a user's next website visit. Dataset includes user's past interactions and demographics. User-based collaborative filtering and popularity with time decay methods improved precision with 3 recommendations by 7% and 8% from a random prediction baseline. Awarded 2nd place at the challenge.
- **Music Recommendation**: Constructed music recommendation system through collaborative filtering based on matrix factoring approach. Fetched and processed ∼12GB of data through `spark`. Utilized `SparkMLib`'s alternating least squares (ALS) algorithm to learn latent factors representations for users and items.
- **Enigma-CDS Datathon**: Exploited various datasets related to New York MTA to identify stations that are the least accessible targeted for tourists, seniors and disabled. Presented the accessibility of each station geographically. Rated the accessibility of each stations by extracting from Wi-Fi, elevator outages, station entrances, and nearby bus stop shelters dataset.